After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

7.1–7.2 · Proteins and amino acids

Q1 CH7-LO1

Classify each amino acid by its side chain.

Options: nonpolar · polar, uncharged · acidic · basic

leucine _____

serine _____

glutamic acid _____

lysine _____

Q2 CH7-LO1

Which amino acid is the only one that is not chiral?

_____

Q3 CH7-LO1

Which is an essential amino acid?

A. lysine
B. glycine
C. alanine
D. serine

Q4 CH7-LO1

Match each protein to its function.

Options: hormone · transport · structural · enzyme

insulin _____

haemoglobin _____

keratin _____

amylase _____

7.3 · Acid–base behaviour

Q5 CH7-LO2

Glycine at pH 1 exists mainly as:

A. $H_3N^+CH_2COOH$ (charge +1)
B. $H_3N^+CH_2COO^-$ (0)
C. $H_2NCH_2COO^-$ (−1)
D. H_2NCH_2COOH

Q6 CH7-LO2

What is the pH called at which an amino acid has no net charge? (two words or its abbreviation)

_____

Q7 CH7-LO2

At pH 6, which way does each amino acid move in electrophoresis?

Options: towards the + electrode · does not move · towards the − electrode

aspartic acid (pI 3.0) _____

alanine (pI 6.0) _____

lysine (pI 9.8) _____

Q8 CH7-LO2

Ninhydrin is used to detect amino acids because it:

A. reacts with the amino group to give a purple colour
B. dissolves proteins
C. is an enzyme
D. measures pH

7.4 · Peptides

Q9 CH7-LO3

In the tripeptide Ala-Gly-Ser, the N-terminal amino acid is:

A. alanine
B. glycine
C. serine
D. it has none

Q10 CH7-LO3

How many water molecules are released when four amino acids join into a tetrapeptide?

_____

Q11 CH7-LO3

Gly-Ala and Ala-Gly are the same compound.

True / False

Q12 CH7-LO3

Sickle cell anaemia results from:

A. one wrong amino acid (Val for Glu) in haemoglobin
B. too little glucose
C. a missing disulfide in insulin
D. a lipid disorder

7.5–7.6 · Proteins, hydrolysis and denaturation

Q13 CH7-LO4

Match each level of structure to what holds it.

Options: peptide bonds · backbone hydrogen bonds · side-chain interactions in one chain · interactions between chains

primary _____

secondary _____

tertiary _____

quaternary _____

Q14 CH7-LO4

Which interaction is covalent?

A. disulfide bridge
B. hydrogen bond
C. ionic bond
D. dispersion forces

Q15 CH7-LO4

Complete hydrolysis of a protein gives:

A. amino acids
B. glucose
C. fatty acids and glycerol
D. nucleotides

Q16 CH7-LO4

70% alcohol disinfects skin because it:

A. denatures microbial proteins
B. hydrolyses DNA
C. is an enzyme
D. is a strong acid

7.7 · Enzymes

Q17 CH7-LO5

Match each enzyme to its class.

Options: hydrolase · oxidoreductase · transferase · isomerase class

lipase _____

alcohol dehydrogenase _____

kinase _____

isomerase _____

Q18 CH7-LO5

The rate of an enzyme reaction levels off at high [substrate] because:

A. all active sites are occupied (saturated)
B. the enzyme is denatured
C. the substrate runs out
D. the product inhibits nothing

Q19 CH7-LO5

A competitive inhibitor:

A. resembles the substrate and binds the active site
B. binds an allosteric site only
C. forms a covalent bond permanently
D. denatures the enzyme

Q20 CH7-LO5

Nerve gases are lethal because they:

A. irreversibly inhibit acetylcholinesterase
B. denature haemoglobin
C. hydrolyse DNA
D. activate amylase

Pharmacy scenario · The diabetic patient's fridge

Scenario

Encik Azman has type 2 diabetes. He injects **insulin**, takes **atorvastatin** and a daily **low-dose aspirin**. He asks why insulin must be kept in the fridge but not frozen, why he cannot take insulin as a tablet, and why his doctor says aspirin "lasts longer than it stays in the blood".

Q21 CH7-LO4

Why must insulin be kept at 2–8 °C and not frozen?

A. Heat or freezing can denature the protein so it no longer works
B. Cold makes it sweeter
C. It is a lipid that melts
D. Freezing hydrolyses it to glucose

Q22 CH7-LO3

Why is insulin injected rather than swallowed?

A. Proteases in the gut would hydrolyse its peptide bonds
B. It is too sweet
C. It is insoluble in water
D. Tablets denature in light

Q23 CH7-LO5

Low-dose aspirin protects against clots for days because it:

A. irreversibly acetylates COX in platelets, which cannot make new enzyme
B. stays in the blood for weeks
C. is a competitive inhibitor
D. denatures platelets

Q24 CH7-LO5

Explain to Encik Azman, in plain words, how atorvastatin works.

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

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PHD125 Pharmaceutical Organic Chemistry · Amino Acids, Proteins and Enzymes

After class: check your understanding

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About 30 minutes. Questions follow the subchapters of the lecture and end with a pharmacy scenario.

Answer key

7.1–7.2 · Proteins and amino acids

Q1 CH7-LO1

Classify each amino acid by its side chain.

leucine → nonpolar; serine → polar, uncharged; glutamic acid → acidic; lysine → basic

FGOB Table 18.3.1.

Q2 CH7-LO1

Which amino acid is the only one that is not chiral?

glycine

R = H, so the α-carbon has two H atoms.

Q3 CH7-LO1

Which is an essential amino acid?

A. lysine

B. glycine

C. alanine

D. serine

His, Ile, Leu, Lys, Met, Phe, Thr, Trp, Val.

Q4 CH7-LO1

Match each protein to its function.

insulin → hormone; haemoglobin → transport; keratin → structural; amylase → enzyme

FGOB Table 18.2.1.

7.3 · Acid–base behaviour

Q5 CH7-LO2

Glycine at pH 1 exists mainly as:

A. $H_3N^+CH_2COOH$ (charge +1)

B. $H_3N^+CH_2COO^-$ (0)

C. $H_2NCH_2COO^-$ (−1)

D. H_2NCH_2COOH

In acid the carboxylate is protonated.

Q6 CH7-LO2

What is the pH called at which an amino acid has no net charge? (two words or its abbreviation)

isoelectric point

pI.

Q7 CH7-LO2

At pH 6, which way does each amino acid move in electrophoresis?

aspartic acid (pI 3.0) → towards the + electrode; alanine (pI 6.0) → does not move; lysine (pI 9.8) → towards the – electrode

Above the pI it is negative; below it, positive.

Q8 CH7-LO2

Ninhydrin is used to detect amino acids because it:

A. reacts with the amino group to give a purple colour

B. dissolves proteins

C. is an enzyme

D. measures pH

Also used for fingerprints (FGOB 18.4).

7.4 · Peptides

Q9 CH7-LO3

In the tripeptide Ala-Gly-Ser, the N-terminal amino acid is:

A. alanine

B. glycine

C. serine

D. it has none

Written N- to C-terminal, left to right.

Q10 CH7-LO3

How many water molecules are released when four amino acids join into a tetrapeptide?

3

One per peptide bond: 3 bonds.

Q11 CH7-LO3

Gly-Ala and Ala-Gly are the same compound.

True / False → **False**

Different N- and C-terminals: they are isomers.

Q12 CH7-LO3

Sickle cell anaemia results from:

A. one wrong amino acid (Val for Glu) in haemoglobin

B. too little glucose

C. a missing disulfide in insulin

D. a lipid disorder

Sequence decides function (FGOB 18.5).

7.5–7.6 · Proteins, hydrolysis and denaturation

Q13 CH7-LO4

Match each level of structure to what holds it.

primary → peptide bonds; secondary → backbone hydrogen bonds; tertiary → side-chain interactions in one chain; quaternary → interactions between chains

FGOB 18.6–18.9.

Q14 CH7-LO4

Which interaction is covalent?

A. disulfide bridge

B. hydrogen bond

C. ionic bond

D. dispersion forces

–S–S– between two cysteines.

Q15 CH7-LO4

Complete hydrolysis of a protein gives:

A. amino acids

B. glucose

C. fatty acids and glycerol

D. nucleotides

All peptide bonds are broken.

Q16 CH7-LO4

70% alcohol disinfects skin because it:

A. denatures microbial proteins

B. hydrolyses DNA

C. is an enzyme

D. is a strong acid

It disrupts the proteins' hydrogen bonds (FGOB 18.10).

7.7 · Enzymes

Q17 CH7-LO5

Match each enzyme to its class.

lipase → hydrolase; alcohol dehydrogenase → oxidoreductase; kinase → transferase; isomerase → isomerase class

FGOB 19.3.

Q18 CH7-LO5

The rate of an enzyme reaction levels off at high [substrate] because:

A. all active sites are occupied (saturated)

B. the enzyme is denatured

C. the substrate runs out

D. the product inhibits nothing

Like a fixed number of taxis (FGOB 19.5).

Q19 CH7-LO5

A competitive inhibitor:

A. resembles the substrate and binds the active site

B. binds an allosteric site only

C. forms a covalent bond permanently

D. denatures the enzyme

More substrate can outcompete it.

Q20 CH7-LO5

Nerve gases are lethal because they:

A. irreversibly inhibit acetylcholinesterase

B. denature haemoglobin

C. hydrolyse DNA

D. activate amylase

They bond to a serine in the active site (FGOB 19.6).

Pharmacy scenario · The diabetic patient's fridge

Q21 CH7-LO4

Why must insulin be kept at 2–8 °C and not frozen?

A. Heat or freezing can denature the protein so it no longer works

B. Cold makes it sweeter

C. It is a lipid that melts

D. Freezing hydrolyses it to glucose

Protein shape must be preserved.

Q22 CH7-LO3

Why is insulin injected rather than swallowed?

A. Proteases in the gut would hydrolyse its peptide bonds

B. It is too sweet

C. It is insoluble in water

D. Tablets denature in light

Most protein drugs are injected.

Q23 CH7-LO5

Low-dose aspirin protects against clots for days because it:

A. irreversibly acetylates COX in platelets, which cannot make new enzyme

B. stays in the blood for weeks

C. is a competitive inhibitor

D. denatures platelets

Irreversible inhibition.

Q24 CH7-LO5

Explain to Encik Azman, in plain words, how atorvastatin works.

His liver makes cholesterol using an enzyme called HMG-CoA reductase. Atorvastatin looks enough like that enzyme's normal substrate to sit in its active site and block it (competitive inhibition), so the liver makes less cholesterol and his LDL ('bad') cholesterol falls. It must be taken regularly because the blocking wears off.

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